Semester VII **B. Tech CSE (AI)**

BCSAI 701 FUZZY LOGIC AND APPLICATION

3L+0T+0P+3C **MM:100**

Unit1: Introduction

Classical Set Theory, Fuzzy set theory, membership function, predicate logic, approximate reasoning

Unit2: Approximate Reasoning

categorical , qualitative, syllogistic reasoning, dispositional reasoning, rule base, linguistic variable, quantifiers

Unit 3: Inference System

characteristics, MAMDANI FIS, functional blocks: database, decision making, fuzzification, defuzzification

Unit 4: Quantification

Database and queries, Fuzzy Events, Means, variances, decision making: types, multi-person, multi-objective, multi-criteria

Unit 5: Control System

Adaptive controller, operational concepts ,neural networks fuzziness.

Text/Reference Books

1. Fuzzy logic with Engineering applications by Timothy J Ross

2. fuzzy Sets and Fuzzy Logic: Theory and Application by George J Klir, Bo Yuan

3. Neural Networks,Fuzzy Logic and Genetic Algorithms : Synthesis and Applications by GA Vijayalakshmi Pai